

A placebo surgery

Arthroscopic knee surgery for arthritis was common, and patients said it helped

Moseley and colleagues (2002) randomly assigned 180 patients with knee osteoarthritis to two real arthroscopic procedures, or to placebo surgery - skin incisions only, with no instruments inserted

Patients, and the staff who assessed their pain, did not know who got which

Over two years, which group reported the most improvement?

None of them.

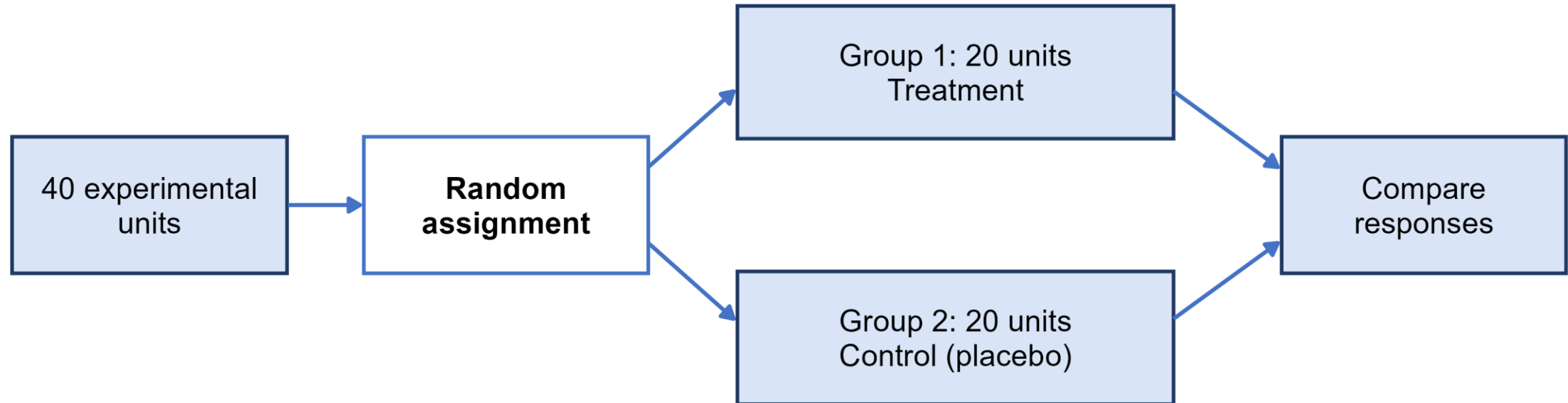
The placebo group did as well as both surgery groups on pain and function. The benefit came from believing you had surgery - and without a placebo group and blinding, nobody would have known.

Source: Moseley et al. (2002), New England Journal of Medicine

Completely randomized design

completely randomized design

every unit is assigned to a treatment completely at random - the simplest design



The Pfizer-BioNTech vaccine trial was a completely randomized design with two groups

Randomized block design

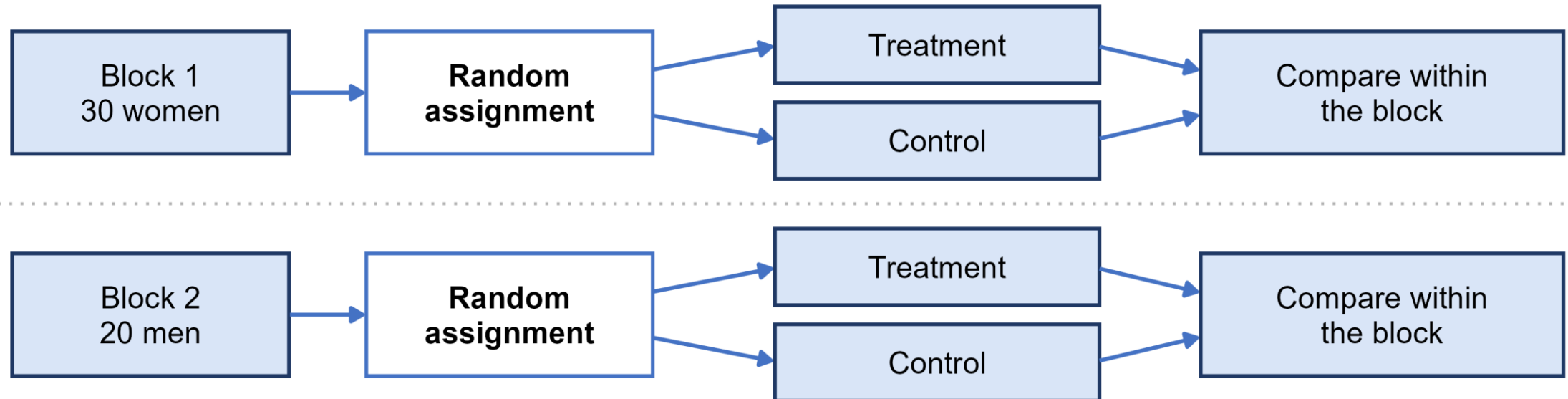
block

a group of units that are alike in a way that affects the response, formed BEFORE treatments are assigned

randomized block design

randomize treatments separately inside each block, then compare treatments within blocks

blocking is to experiments what stratifying is to sampling

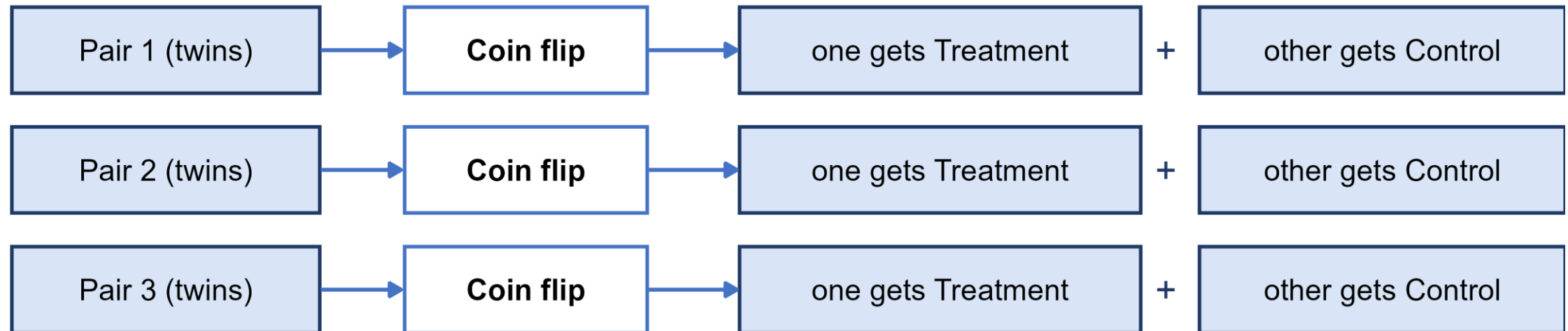


Matched pairs design

matched pairs design

a block design where every block is a pair of very similar units, and a coin flip decides which one gets each treatment

or each subject is their own pair, receiving both treatments in a random order



Before-and-after data with no random order or control group is much weaker evidence

What does randomization buy you?

Random sampling lets you generalize from the sample to the population

Random assignment lets you conclude cause and effect

Most studies have only one. A trial on volunteers shows a vaccine works - for people like those volunteers

		Randomly ASSIGNED to treatments?	
		Yes	No
Randomly SAMPLED?	Yes	Cause, AND generalize to the population rare: an experiment on a random sample	Generalize, but not cause e.g. a random-sample survey
	No	Cause, but only for units like these e.g. most clinical trials, which use volunteers	Association in these units only e.g. volunteers who chose their own treatment

Ethics: the Facebook experiment

For one week in January 2012, Facebook changed the News Feeds of 689,003 users: some saw fewer positive posts, some fewer negative ones. The results were published in 2014. The users never knew.

1. Facebook said agreeing to its Data Use Policy counted as informed consent. Do you agree?

The journal did not: its Expression of Concern said the study may not have met the principles of informed consent and the chance to opt out.

2. Why did no ethics board stop it?

Facebook, a private company, was not bound by the federal rules. Cornell's board decided its researcher was not directly involved, so it did no review.

3. What is the lesson?

Clicking I agree on a terms-of-service page is not informed consent - even for a well-designed experiment.

Sources: Kramer, Guillory and Hancock (2014), PNAS; PNAS Editorial Expression of Concern (2014)

The Belmont Report: three principles

Why the rules exist: in the U.S. Public Health Service syphilis study at Tuskegee (1932 to 1972), hundreds of Black men with syphilis were left untreated without informed consent. The Belmont Report (1979) followed.

respect for persons

people decide for themselves whether to take part

beneficence

do no harm: maximize the possible benefits and minimize the possible risks

justice

the burdens and benefits of research are shared fairly

Research ethics today

informed consent

subjects are told the purpose, procedures and risks in advance, and can refuse or quit at any time

confidentiality

information about individuals is protected and never released in a form that identifies them

Institutional Review Board (IRB)

reviews and approves studies on human subjects before they start
at UI, through the Office of Research Assurances

Ethics: honest data

Ethics also means honest data. An investigation of psychologist Diederik Stapel, infamous social psychologist, found he had invented data: it tainted more than 55 of his papers and 10 PhD dissertations he supervised. A researcher is surveying households in a community:

Source: Introductory Statistics 2e (OpenStax), Section 1.4 and Example 1.22

Practice: what design, and can it show cause?

1. A survey finds that people who own dogs have lower blood pressure than people who do not.
Observational - nobody was assigned a dog. Active people may both own dogs and have lower blood pressure.
2. 60 volunteers with insomnia are randomly split: 30 get melatonin, 30 get an identical-looking placebo.
Completely randomized experiment - it can show cause, for people like these volunteers.
3. In each of 4 greenhouses, tomato plants are randomly assigned to fertilizer A or fertilizer B.
Randomized block design, with greenhouses as the blocks - it can show cause.

What you should know: studies and designs

Three kinds of study - be ready to define and identify each one

An experimental study assigns the treatments. An observational study only watches and measures. A survey samples a population in order to describe it, and assigns nothing.

The sampling designs

Simple random, systematic, stratified, cluster, and two-stage cluster. Convenience and voluntary response samples are not random, and they are biased.

The experimental designs

Completely randomized, randomized block, and matched pairs.

What you should know: terminology

The language of an experiment

Experimental units, the factor, its levels, the treatments, the control group, a placebo, blinding and double blinding.

Association is not causation

A lurking variable is associated with both the explanatory and the response variable, and can produce an association where there is no causal link.

The two things randomness buys you

Random sampling lets you generalize to the population. Random assignment lets you claim cause. A bigger sample shrinks sampling error, but it never removes bias.